

HACKPAD – 4–key macropad + encoder + OLED

MCU: Seeed XIAO RP2040 (through–hole; 2x 1x7 socket headers)
Inputs: 4 MX switches (2x2) + 1 EC11 rotary encoder w/ push (= 5 inputs)
Display: 0.91" SSD1306 128x32 OLED, I2C1 @ 0x3C (pin order: GND–VCC–SCL–SDA)
Optional: SK6812 RGB chain off XIAO D10 (hand–wired, not on PCB)

Pin map:

D0	GP26	COL0	D6	GP0	ROW0
D1	GP27	COL1	D7	GP1	ROW1
D2	GP28	ENC_A	D8	GP2	ENC_PUSH
D3	GP29	ENC_B	D10	GP3	RGB_DIN (optional)
D4	GP6	I2C SDA	D5	GP7	I2C SCL

*This board is PCB–first: the .kicad_pcb file is authoritative.
All switches, diodes, XIAO sockets, OLED header, encoder, mounting holes,
edge cuts, COL/ROW trunks, switch–anode bridges, GND pour are routed.*

Seeed XIAO RP2040 (KMK firmware) – see pcb/netlist.csv 0.91" 128x32 SSD1306 OLED on I2C1 EC11 rotary encoder with push–switch 4 MX switches in a 2x2 COL2ROW matrix Hackpad		
Sheet: / File: hackpad.kicad_sch		
Title: Hackpad – 4–key + Encoder Macropad		
Size: A4	Date: 2026–06–07	Rev: 1.0
KiCad E.D.A. 9.0.8		Id: 1/1